

HDOC Day-21

Menu-Driven C Program Using Nested Loops and Switch-Case

Problem Statement

Write a menu-driven C program that accepts two positive integers m and n , where $m \leq n$. Use a switch statement to repeatedly perform the selected operation until Exit is chosen: (1) first 10 multiples, (2) addition tables using 1 to 10, (3) subtraction tables using 1 to 10, (4) division tables using 1 to 10 with results rounded to two decimal places, and (5) exit. Display an error for invalid menu choices.

Algorithm

1. Start.
2. Input m and n .
3. Validate that m and n are positive and $m \leq n$. If not, display an error and ask again.
4. Use a do-while loop to display the menu until choice 5 is selected.
5. Input the menu choice and use switch-case.
6. Choice 1: For each integer i from m to n , use an inner loop $j = 1$ to 10 and print $i * j$.
7. Choice 2: For each integer i from m to n , use an inner loop $j = 1$ to 10 and print $i + j$.
8. Choice 3: For each integer i from m to n , use an inner loop $j = 1$ to 10 and print $i - j$.
9. Choice 4: For each integer i from m to n , use an inner loop $j = 1$ to 10 and print $(\text{double})i / j$ to two decimal places.
10. Choice 5: Display an exit message and terminate the loop.
11. Default: Display an invalid-choice message.
12. Stop.

Test Case Table

TC	m	n	Choice	Purpose / Expected Result
1	2	3	1	Valid: prints 2 and 3 multiplication tables (x1 to x10).
2	2	3	2	Valid: prints addition tables for 2 and 3 using +1 to +10.
3	2	3	3	Valid: prints subtraction tables for 2 and 3 using -1 to -10.
4	2	3	4	Valid: prints division results for 2 and 3 by 1 to 10, to 2 decimals.
5	5	5	1	Boundary: $m = n$; prints only the table for 5.
6	0	3	—	Invalid: m is not positive; asks for m and n again.
7	5	2	—	Invalid: $m > n$; asks for m and n again.
8	2	3	9	Invalid menu choice; displays an error and menu continues.
9	2	3	5	Exit: program terminates normally.

C Program

```
#include <stdio.h>

int main(void)
{
    int m, n, choice;
    int i, j;

    /* Read valid values of m and n. */
    do
    {
        printf("Enter two positive integers m and n (m <= n): ");
        scanf("%d %d", &m, &n);

        if (m <= 0 || n <= 0 || m > n)
        {
            printf("Invalid input. Enter positive integers with m <= n.\n\n");
        }
    }
    while (m <= 0 || n <= 0 || m > n);

    do
    {
        printf("\nMenu:\n");
        printf("1. First 10 multiples\n");
        printf("2. Addition tables (1 to 10)\n");
        printf("3. Subtraction tables (1 to 10)\n");
        printf("4. Division tables (1 to 10, 2 decimals)\n");
        printf("5. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                for (i = m; i <= n; i++)
                {
                    for (j = 1; j <= 10; j++)
                        printf("%d * %d = %d\n", i, j, i * j);
                }
                break;
            case 2:
                for (i = m; i <= n; i++)
                {
                    for (j = 1; j <= 10; j++)
                        printf("%d + %d = %d\n", i, j, i + j);
                }
                break;
            case 3:
                for (i = m; i <= n; i++)
                {
                    for (j = 1; j <= 10; j++)
                        printf("%d - %d = %d\n", i, j, i - j);
                }
                break;
            case 4:
                for (i = m; i <= n; i++)
                {
                    for (j = 1; j <= 10; j++)
                        printf("%d / %d = %.2f\n", i, j, (double)i / j);
                }
                break;
            case 5:
                printf("Exit\n");
                break;
            default:
                printf("Invalid choice. Please enter a valid choice.\n\n");
        }
    }
    while (choice != 5);

    printf("Program terminated successfully.\n");
    return 0;
}
```

```

    }
} while (m <= 0 || n <= 0 || m > n);

do
{
    printf("\n===== MENU =====\n");
    printf("1. Print first 10 multiples\n");
    printf("2. Print addition tables\n");
    printf("3. Print subtraction tables\n");
    printf("4. Print division tables\n");
    printf("5. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            for (i = m; i <= n; i++)
            {
                printf("\nMultiples of %d:\n", i);
                for (j = 1; j <= 10; j++)
                    printf("%d x %d = %d\n", i, j, i * j);
            }
            break;

        case 2:
            for (i = m; i <= n; i++)
            {
                printf("\nAddition table of %d:\n", i);
                for (j = 1; j <= 10; j++)
                    printf("%d + %d = %d\n", i, j, i + j);
            }
            break;

        case 3:
            for (i = m; i <= n; i++)
            {
                printf("\nSubtraction table of %d:\n", i);
                for (j = 1; j <= 10; j++)
                    printf("%d - %d = %d\n", i, j, i - j);
            }
            break;

        case 4:
            for (i = m; i <= n; i++)
            {
                printf("\nDivision table of %d:\n", i);
                for (j = 1; j <= 10; j++)
                    printf("%d / %d = %.2f\n", i, j, (double)i / j);
            }
            break;

        case 5:
            printf("Exiting program.\n");
            break;

        default:
            printf("Invalid menu choice. Please enter 1 to 5.\n");
    }
} while (choice != 5);

return 0;
}

```

Compilation and Execution

Compile: gcc hdoc21.c -o hdoc21

Run: ./hdoc21

Screenshot 1 — Source Code / Compilation

```
#include <stdio.h>
int main(void)
{
    int m, n, choice;
    int i, j;
    /* Read valid values of m and n. */
    do
    {
        printf("Enter two positive integers m and n ( $m \leq n$ ): ");
        scanf("%d %d", &m, &n);
        if (m ≤ 0 || n ≤ 0 || m > n)
        {
            printf("Invalid input. Enter positive integers with  $m \leq n$ .\n\n");
        }
    } while (m ≤ 0 || n ≤ 0 || m > n);
    do
    {
        printf("\n===== MENU =====\n");
        printf("1. Print first 10 multiples\n");
        printf("2. Print addition tables\n");
        printf("3. Print subtraction tables\n");
        printf("4. Print division tables\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice)
        {
            case 1:
```

```
            case 1:
                for (i = m; i ≤ n; i++)
                {
                    printf("\nMultiples of %d:\n", i);
                    for (j = 1; j ≤ 10; j++)
                    {
                        printf("%d x %d = %d\n", i, j, i * j);
                    }
                    break;
                }
            case 2:
                for (i = m; i ≤ n; i++)
                {
                    printf("\nAddition table of %d:\n", i);
                    for (j = 1; j ≤ 10; j++)
                    {
                        printf("%d + %d = %d\n", i, j, i + j);
                    }
                    break;
                }
            case 3:
                for (i = m; i ≤ n; i++)
                {
                    printf("\nSubtraction table of %d:\n", i);
                    for (j = 1; j ≤ 10; j++)
                    {
                        printf("%d - %d = %d\n", i, j, i - j);
                    }
                    break;
                }
            case 4:
                for (i = m; i ≤ n; i++)
                {
                    printf("\nDivision table of %d:\n", i);
```

```

for (j = 1; j ≤ 10; j++)
printf("%d / %d = %.2f\n", i, j, (double)i / j);
}
break;
case 5:
printf("Exiting program.\n");
break;
default:
printf("Invalid menu choice. Please enter 1 to 5.\n");
}
while (choice ≠ 5);
return 0;
}

```

```

(kali@kali)-[~/Desktop]
$ vi hdoc21.c

```

```

(kali@kali)-[~/Desktop]
$ gcc hdoc21.c -o hdoc21

```

```

(kali@kali)-[~/Desktop]
$ ./hdoc21
Enter two positive integers m and n (m ≤ n): 2 3

===== MENU =====
1. Print first 10 multiples
2. Print addition tables
3. Print subtraction tables
4. Print division tables
5. Exit
Enter your choice: 1

Multiples of 2:
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20

```

```

Multiples of 3:
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30

```

```
4. Print division table
5. Exit
Enter your choice: 2

Addition table of 2:
2 + 1 = 3
2 + 2 = 4
2 + 3 = 5
2 + 4 = 6
2 + 5 = 7
2 + 6 = 8
2 + 7 = 9
2 + 8 = 10
2 + 9 = 11
2 + 10 = 12
```

```
Addition table of 3:
3 + 1 = 4
3 + 2 = 5
3 + 3 = 6
3 + 4 = 7
3 + 5 = 8
3 + 6 = 9
3 + 7 = 10
3 + 8 = 11
3 + 9 = 12
3 + 10 = 13
```

```
4. Print division tables
5. Exit
Enter your choice: 3

Subtraction table of 2:
2 - 1 = 1
2 - 2 = 0
2 - 3 = -1
2 - 4 = -2
2 - 5 = -3
2 - 6 = -4
2 - 7 = -5
2 - 8 = -6
2 - 9 = -7
2 - 10 = -8

Subtraction table of 3:
3 - 1 = 2
3 - 2 = 1
3 - 3 = 0
3 - 4 = -1
3 - 5 = -2
3 - 6 = -3
3 - 7 = -4
3 - 8 = -5
3 - 9 = -6
3 - 10 = -7
```

```
5. Exit
Enter your choice: 4

Division table of 2:
2 / 1 = 2.00
2 / 2 = 1.00
2 / 3 = 0.67
2 / 4 = 0.50
2 / 5 = 0.40
2 / 6 = 0.33
2 / 7 = 0.29
2 / 8 = 0.25
2 / 9 = 0.22
2 / 10 = 0.20
```

```
Division table of 3:
3 / 1 = 3.00
3 / 2 = 1.50
3 / 3 = 1.00
3 / 4 = 0.75
3 / 5 = 0.60
3 / 6 = 0.50
3 / 7 = 0.43
3 / 8 = 0.38
3 / 9 = 0.33
3 / 10 = 0.30
```

```
(kali@kali)-[~/Desktop]
$ ./hdoc21
Enter two positive integers m and n ( $m \leq n$ ): 5 5

===== MENU =====
1. Print first 10 multiples
2. Print addition tables
3. Print subtraction tables
4. Print division tables
5. Exit
Enter your choice: 1

Multiples of 5:
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

Screenshot 2 — Invalid Choice & Exit

```
===== MENU =====
1. Print first 10 multiples
2. Print addition tables
3. Print subtraction tables
4. Print division tables
5. Exit
Enter your choice: 9
Invalid menu choice. Please enter 1 to 5.
```

```
(kali㉿kali)-[~/Desktop]  
$ ./hdoc21
```

```
Enter two positive integers m and n ( $m \leq n$ ): 0 3  
Invalid input. Enter positive integers with  $m \leq n$ .
```

```
Enter two positive integers m and n ( $m \leq n$ ): 5 2  
Invalid input. Enter positive integers with  $m \leq n$ .
```

```
===== MENU =====  
1. Print first 10 multiples  
2. Print addition tables  
3. Print subtraction tables  
4. Print division tables  
5. Exit  
Enter your choice: 5  
Exiting program.
```